

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _Narmatha J (192521341)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _NARMATHA J_

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. Auto-Scaling and Load Balancing for Seasonal Sales

A company may experience a sudden increase in users during seasonal sales such as Amazon Great Indian Festival, Flipkart Big Billion Days, or Black Friday. Cloud computing can handle this problem using auto-scaling and load balancing.

Auto-Scaling:

Auto-scaling automatically increases or decreases the number of cloud servers according to traffic.

- During normal traffic, fewer servers are used.
- During a sale, traffic increases suddenly.
- Auto-scaling automatically adds more servers.
- After the sale, unnecessary servers are removed.
- This saves cost because the company pays for resources based on demand.

Load Balancing:

A load balancer distributes incoming user requests across multiple servers.

For example:

Users → Load Balancer → Server 1 / Server 2 / Server 3 / Server 4
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If one server becomes overloaded, requests can be redirected to other available servers.

Benefits:

- Prevents server overload.
- Reduces downtime.
- Improves application response time.
- Provides better availability.

- Handles sudden traffic spikes efficiently.
- Improves customer experience.

Role of Monitoring Tools:

Monitoring tools continuously check:

- CPU and memory usage
- Number of users
- Network traffic
- Server health
- Application response time
- Error rates

When traffic reaches a particular threshold, monitoring systems can trigger auto-scaling.

Real-World Example:

During an **Amazon festival sale**, millions of customers may simultaneously search for products and place orders. Auto-scaling adds servers during high traffic, while load balancing distributes requests among them. Monitoring tools identify traffic increases and server failures.

Result: The website remains available, responds faster, and reduces the possibility of crashes during the sale.

2. Secure Cloud Storage for Backup Data

Organizations generate large amounts of backup data such as databases, documents, application files, and transaction records. Cloud storage provides a secure and scalable method to store this information.

Durability and Data Redundancy:

Cloud storage providers store multiple copies of data in different storage devices or locations.

Example:

Original Data → Copy 1 + Copy 2 + Copy 3

If one storage device fails, another copy can be used.

This provides:

- High durability
- Data redundancy
- Fault tolerance
- Data recovery
- Protection against hardware failure

Encryption:

Encryption converts readable data into an unreadable format.

Two important types are:

- **Encryption at rest:** Protects stored backup data.
- **Encryption in transit:** Protects data while it is being transferred.

Only authorized users with the required keys can access encrypted data.

Access Control:

Access control ensures that only authorized people can access backup data.

It can use:

- Usernames and passwords
- Multi-factor authentication
- Role-based access control
- Permissions
- Encryption keys

For example, only the database administrator may be allowed to restore company backups.

Cost Advantages:

Cloud storage can be cheaper than traditional storage because:

- No need to purchase large physical storage systems.
- No separate data-center infrastructure is required.
- Reduced hardware maintenance costs.
- Storage capacity can be increased when required.
- Pay-as-you-use pricing is available.
- Older backup data can be moved to cheaper storage tiers.

Disaster Recovery Example:

Suppose a company's main data center is destroyed due to a **fire or flood**.

The company can recover its applications and databases from cloud backups stored in another region.

Primary Data Center → Cloud Backup → Disaster → Restore Data → New/Secondary Data Center

Result:

Cloud storage provides durability, redundancy, security, scalability, and cost savings, making it suitable for disaster recovery and backup management.

3. CDN and Global Application Deployment

A company that provides a mobile application globally may have users from India, the USA, Europe, and other regions. If all users access a single server location, users far away may experience higher latency.

A **Content Delivery Network (CDN)** solves this problem.

How CDN Works?

A CDN stores frequently accessed content such as:

- Images
- Videos
- JavaScript files
- CSS files
- Application resources

in multiple geographically distributed **edge servers**.

User → Nearest Edge Server → Content

Instead of:

User → Distant Central Server → Content

Benefits of CDN:

- Reduces latency.
- Improves loading speed.
- Reduces traffic to the main server.
- Improves application performance.
- Handles large numbers of users.
- Improves availability.

Role of Edge Servers:

Edge servers are located closer to users.

For example:

- Indian user → Mumbai/Chennai edge location
- European user → European edge location
- US user → US edge location

The content travels a shorter distance, so the user receives it faster.

Global Cloud Availability:

Cloud providers support global applications using:

- Multiple regions
- Availability zones

- Global load balancing
- CDN services
- Replicated databases
- Disaster recovery systems

If one region becomes unavailable, traffic can potentially be redirected to another region.

Real-World Example – Netflix:

Consider a video streaming application such as **Netflix**.

If a user in India requests a popular movie, the video content can be delivered from a nearby CDN/edge location instead of directly from a distant central server.

This provides:

- Faster video startup
- Reduced buffering
- Lower latency
- Better streaming quality

Result:

CDNs and edge servers make global applications **faster, more reliable, and responsive**, especially for video streaming, online gaming, and other real-time applications.

4. Cloud Security and Compliance for Financial Institutions

Financial institutions handle highly sensitive information such as bank account details, credit card information, customer identity data, and financial transactions. Therefore, strong cloud security and compliance mechanisms are required.

Regulatory Compliance:

Cloud providers provide security services and certifications that help organizations meet regulatory requirements.

Organizations can use:

- Compliance frameworks
- Security policies
- Audit reports
- Identity management
- Data encryption
- Security monitoring

The company must still configure and operate its cloud environment correctly; compliance is generally a shared responsibility.

Logging:

Logging records activities occurring within the cloud environment.

For example:

User → Login → Access Account → Transfer Money → Logout

Logs can record:

- Login attempts
- Data access
- Transaction activities
- Configuration changes
- Failed authentication attempts

These records help during security investigations and audits.

Monitoring:

Monitoring continuously checks the cloud environment for suspicious activity.

It can detect:

- Unusual login locations
- Multiple failed login attempts
- Unexpected data access
- Network attacks
- Abnormal transaction activity

Encryption:

Encryption protects sensitive financial information.

For example:

Customer Account Data → Encryption → Secure Cloud Storage

Even if unauthorized people obtain the stored data, encrypted information is difficult to understand without the required key.

Data Sovereignty:

Data sovereignty means that an organization must comply with the laws and regulations of the country where its data is stored or processed.

A financial institution can:

- Select appropriate cloud regions.
- Store sensitive information in permitted countries.
- Apply location-based access policies.
- Maintain records of where data is stored.
- Use appropriate encryption and compliance controls.

Example – Online Banking:

Suppose a customer transfers **₹50,000** using online banking.

The cloud system can:

1. Authenticate the customer.
2. Encrypt the transaction.
3. Record the transaction in logs.
4. Monitor for suspicious behavior.
5. Store transaction information securely.
6. Generate audit records.

Result:

Cloud security mechanisms such as encryption, logging, monitoring, access control, and compliance auditing help financial institutions protect sensitive transactions and meet regulatory requirements.

5. Containers, Virtual Machines and Kubernetes

Containerization allows developers to package an application together with its dependencies so that it can run consistently across different environments.

Containers vs Virtual Machines:

Feature	Containers	Virtual Machines
Operating System	Share host OS kernel	Each VM has its own OS
Resource Usage	Low	Higher
Startup Time	Very fast	Slower
Performance	Near-native	More overhead
Size	Lightweight	Larger
Deployment	Fast	Comparatively slower
Scalability	Easy	Requires more resources

Container Architecture:

Application + Libraries + Dependencies → Container

Multiple containers can run on the same host while sharing the host operating system kernel.

Kubernetes Orchestration:

When an application contains hundreds of containers, manually managing them becomes difficult.

Kubernetes automates container management.

It can:

- Deploy containers.
- Start and stop containers.
- Scale applications.
- Restart failed containers.
- Distribute workloads.
- Perform service discovery.
- Support rolling updates.

Scaling Example:

Suppose an application normally requires **3 containers**.

During high traffic:

3 Containers → Kubernetes → 10 Containers

After traffic decreases:

10 Containers → Kubernetes → 3 Containers

This improves resource utilization and application availability.

Portability:

Containers provide portability because the same container image can run across different environments.

For example:

Developer Laptop → Private Cloud → AWS → Azure → Google Cloud

The application does not need to be completely rewritten for each environment.

Microservices Example:

Consider an **online shopping application**.

It can be divided into separate microservices:

E-Commerce Application:

- User Service
- Product Service
- Shopping Cart Service
- Payment Service
- Order Service
- Notification Service

Each service can run inside its own container.

User → Load Balancer → Kubernetes → Microservices

If the payment service receives heavy traffic, Kubernetes can scale only the payment service rather than the entire application.

Result:

Containers provide lightweight, fast, and portable deployment, while Kubernetes provides automated management, scaling, recovery, and orchestration of containerized applications.